Hints: Polynomials – 4. Real roots: Descartes, Sturm and Newton's inequalities

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $V(x)$ can only change when x passes a root of some P_k. Check two facts: consecutive P_k, P_{k+1} have no common root (else, going back through $P_{k-1} = q_k P_k - P_{k+1}$, it would be a common root of P and P'), and $P_{k-1}(c) = -P_{k+1}(c)$ when $P_k(c) = 0$.

Hint 2. Hence passing a root of an intermediate P_k does not change the number of sign changes in P_{k-1}, P_k, P_{k+1}. Passing a root c of P: near c, P/P' goes from negative to positive, so exactly one sign change is lost. This gives $V(a) - V(b)$ roots in $(a, b]$.

Hint 3. For $x^5 - 3x - 1$ the sequence is $x^5 - 3x - 1,\ 5x^4 - 3,\ \frac{12}{5}x + 1,\ c$ with a positive constant c: $V(-\infty) = 3$, $V(+\infty) = 0$, so 3 real roots. For the second polynomial, $x^4 + x^3 - x - 1 = (x+1)(x-1)(x^2+x+1)$ has 2 real roots; check that Sturm agrees.

Problem 2

Hint 1. Differentiate: $f'(x) = 1 + x + \cdots + x^{n-1} = \frac{x^n - 1}{x - 1}$. Where does f' vanish? Distinguish n even and n odd.

Hint 2. For n odd, $f' > 0$ everywhere, so f is increasing and has exactly one real root. For n even, f' vanishes only at $x = -1$, which is the global minimum of f.

Hint 3. $f(-1) = \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \cdots + \frac{1}{n} > 0$ (alternating, decreasing terms). Answer: no real roots for n even, exactly one for n odd.

Problem 3

Hint 1. Let $x_1 < x_2 < x_3$ be the roots of $f(x) - s$. Vieta gives $x_1 + x_2 + x_3 = \alpha + \beta$ and $x_1x_2 + x_1x_3 + x_2x_3 = \alpha\beta$, independently of s. Express x_1x_3 through the middle root $m = x_2$.

Hint 2. $x_1x_3 = \alpha\beta - m(\alpha + \beta - m) = \left(m - \frac{\alpha+\beta}{2}\right)^2 + \alpha\beta - \frac{(\alpha+\beta)^2}{4}$.

Hint 3. As s runs over S, the middle root runs over the open interval between the critical points of f. Check that $\frac{\alpha+\beta}{2}$ lies in it (this uses $\alpha\beta < 0$). Minimum: $-\frac{(\beta-\alpha)^2}{4}$.

Problem 4

Hint 1. Translate the divisibility into linear conditions: $p^{(i)}(1) = 0$ for $i \leqslant k$ means $\sum_{j=0}^{n-1} a_j g(j) = -g(n)$ for every polynomial g of degree $\leqslant k$.

Hint 2. Hence $\sum |a_j| \geqslant \dfrac{|g(n)|}{\max_{0 \leqslant j \leqslant n-1} |g(j)|}$. Choose g small on $[0, n-1]$ but large at n: a Chebyshev polynomial. (Note $k + 1 \leqslant n$.)

Hint 3. Take $g(x) = T_k\left(\frac{2x-(n-1)}{n-1}\right)$, so $|g| \leqslant 1$ on $[0, n-1]$ and $g(n) = T_k\left(1 + \frac{2}{n-1}\right)$. Since $T_k(1) = 1$, $T_k'(1) = k^2$ and T_k is convex on $[1, \infty)$, $g(n) \geqslant 1 + \frac{2k^2}{n-1} > 1 + \frac{2k^2}{n}$.

Problem 5

Hint 1. Subtract the line: $Q = P - \ell$ has degree 5, $Q'' = P''$, and Q vanishes at the three inflection abscissae $x_1 < x_2 < x_3$, which are also the roots of Q''.

Hint 2. Normalize $x_2 = 0$ and the leading coefficient to 1: $Q'' = 20(x + a)x(x - b)$. Integrate twice and impose $Q(-a) = Q(0) = Q(b) = 0$; you should find that $a = b$ is forced (for $a, b > 0$).

Hint 3. With $a = b = 1$: $Q = x^5 - \frac{10}{3}x^3 + \frac{7}{3}x = x(x^2-1)(x^2-\frac{7}{3})$, odd, with roots $0, \pm 1, \pm\sqrt{7/3}$. There are four bounded regions; affine maps preserve ratios of areas.

Hint 4. The areas are $\int_0^1 Q = \frac{1}{2}$ and $\int_1^{\sqrt{7/3}} |Q| = \frac{16}{81}$. So the areas, in order along the line, are in ratio $32 : 81 : 81 : 32$.

Problem 6

(a)

Hint 1. Subtract the line: $P - \ell$ has degree 6 and double roots at the three abscissae, so $P - \ell = c(x - x_1)^2(x - x_2)^2(x - x_3)^2$.

Hint 2. Ratios of lengths on the line equal ratios of the abscissae differences. With $x_2 = 0$ and $x_1 = -x_3$ the polynomial is even, so the two areas are equal.

(b)

Hint 1. Put $x_2 = 0$, $x_1 = -1$, $x_3 = k$ and compute $\int_0^k x^2(x + 1)^2(x - k)^2 dx$ and $\int_{-1}^0$ using $\int_0^b x^m(b - x)^2 dx$.

Hint 2. $\int_0^k x^2(k - x)^2(x + 1)^2 dx = k^5(\frac{k^2}{105} + \frac{k}{30} + \frac{1}{30})$, and the other area is $\frac{1}{105} + \frac{k}{30} + \frac{k^2}{30}$.

Hint 3. So $K = k^5 \dfrac{2k^2 + 7k + 7}{7k^2 + 7k + 2}$; show that the fraction lies strictly between $\frac{2}{7}$ and $\frac{7}{2}$ for $k > 0$.

Problem 7

Hint 1. Multiply P by $(1 - z)$ (or $(z - 1)$) and use the monotonicity of the coefficients.

Hint 2. If $P(z) = 0$, then $(1 - z)P(z) = 0$ gives $a_n = a_0 z^{n+1} + \sum_{k=1}^{n}(a_k - a_{k-1})z^{n+1-k}$, where all the coefficients a_0, $a_k - a_{k-1}$ are positive with sum a_n.

Hint 3. For $|z| \leqslant 1$ the right-hand side has modulus $\leqslant a_n$, with equality only if all the powers z^j equal 1, i.e. $z = 1$; but $P(1) > 0$.

Problem 8

Hint 1. Write $p(x) = c\prod_i(1 - x/z_i)$ with $|1/z_i| < 1$. A geometric-series trick turns each factor into one with tiny non-constant coefficients.

Hint 2. $(1-x/z)(1+x/z+\cdots+(x/z)^{N-1}) = 1-(x/z)^N$. Take $q(x) = \prod_i(1-x^N/z_i^N)$ (product over all roots with multiplicity); it is real because the roots come in conjugate pairs, and $p \mid q$.

Hint 3. q has constant term 1, and the sum of absolute values of its other coefficients is at most $\prod_i(1 + |z_i|^{-N}) - 1 \to 0$ as $N \to \infty$.

Problem 9

Hint 1. If two roots of P lay between consecutive roots of P', Rolle would give a root of P' between them.

Hint 2. For $c \neq 0$: $P + cP' = c\,e^{-x/c}\big(e^{x/c}P\big)'$. Apply Rolle to $e^{x/c}P$ between consecutive distinct roots of P, and account for multiple roots (m-fold root of P is an $(m-1)$-fold root of $P + cP'$).

Hint 3. This gives $n-1$ roots; the last one comes from the end where $e^{x/c}P \to 0$ (left of the smallest root if $c > 0$, right of the largest if $c < 0$). All n roots are real and interlace with those of P.

Problem 10

Hint 1. Write all the exponents with a common denominator N and put $t = e^{x/N} > 0$. The equation becomes $g(t) = \sum t^{A_i} - \sum t^{B_i} = 0$ with integers A_i, B_i, and $t = 1$ is always a root.

Hint 2. Divide by $t - 1$: the coefficients of $h(t) = g(t)/(t-1)$ are, up to sign, the partial sums of the coefficients of g, i.e. $\#\{i : A_i \leqslant m\} - \#\{i : B_i \leqslant m\}$ at position m.

Hint 3. This sequence of partial sums starts and ends at 0 and moves in steps ± 1 at the $2n$ exponents; each excursion away from 0 uses at least two exponents, so it has at most $n-1$ sign changes. Descartes for h gives at most $n-1$ positive roots, plus $t = 1$: at most n roots.